Variant Effect Predictor (VEP)

A Practical Introduction

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Somatic vs Germline

- A germline mutation is inherited by the individual from birth. They can be oncogenic (ie Rb = retinoblastoma)
- A somatic mutation, or acquired mutation, happens in somatic cells instead of germ cells and will not pass to offspring (ie TP53).

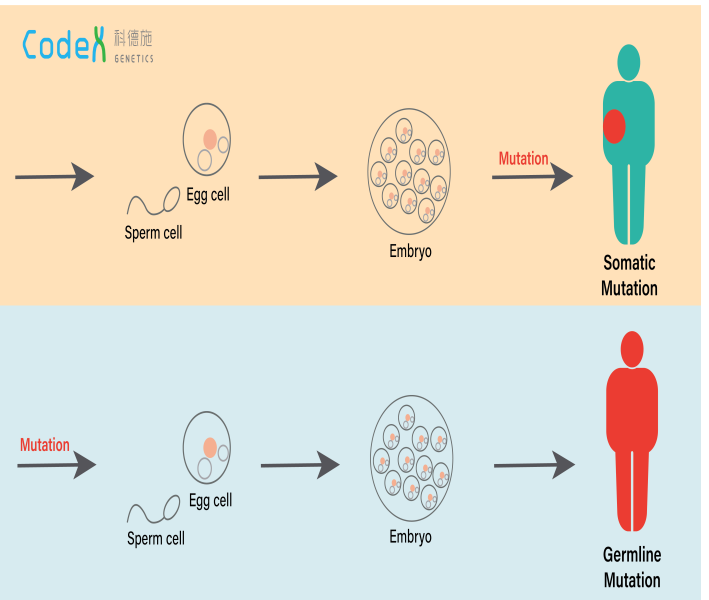
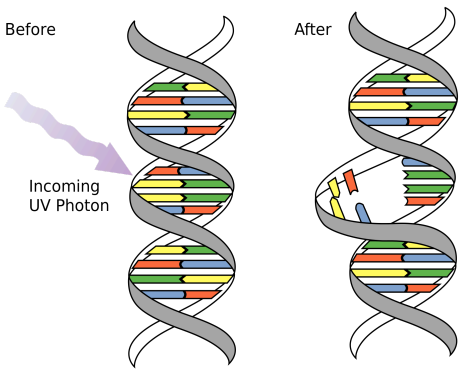


Figure 1: Germline vs somatic

How do mutations appear?

- Mutations occur due to replication errors (1 nucleotide per ~10^4) but with reparation mechanisms (1 error per 10^7 - 10^9)
- As a consequence of DNA damage (~70,000 nucleotide lesions or modifications per day)
- Exposure to mutagens such as UV, smoking increases the mutations' frequency



Example UV damage,knowgenetics.org

Can the cell repair them?

Yes, there are many mechanisms of DNA repair. Some of the most common are:

- **NER (Nucleotide Excision Repair)**: Repairs bulky lesions such as thymine dimers caused by UV radiation. This mechanism involves the removal of a short single-stranded DNA segment containing the damage, followed by DNA synthesis using the complementary strand as a template.
- **MMR (Mismatch Repair)**: Fixes replication errors such as base mismatches or insertion/deletion loops. MMR recognizes the newly synthesized strand and corrects errors by removing the incorrect nucleotides and replacing them.
- **BER (Base Excision Repair)**: Repairs single-base lesions such as oxidative damage, alkylation, or deamination. This involves the removal of damaged bases by specific glycosylases, followed by the excision of the resulting abasic site.
- **HR (Homologous Recombination)**: Repairs double-strand breaks using a homologous sequence as a template, typically from a sister chromatid. This is an error-free repair mechanism.
- **NHEJ (Non-Homologous End Joining)**: Repairs double-strand breaks without the need for a homologous template. While quicker, this method is error-prone and can lead to insertions or deletions.

Visualizing DNA Repair Mechanisms

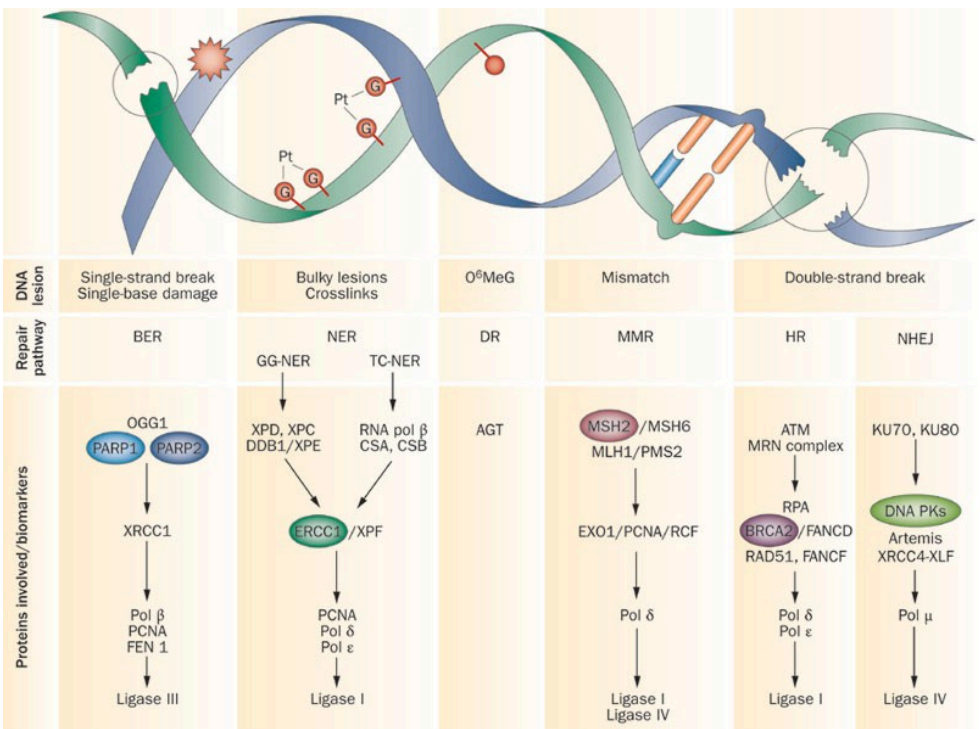


Figure 2: DNA Repair Mechanisms, The origin of human mutation in light of genomic data, Seplyarskiy et al, 2021

Acquisition of mutation in cancer

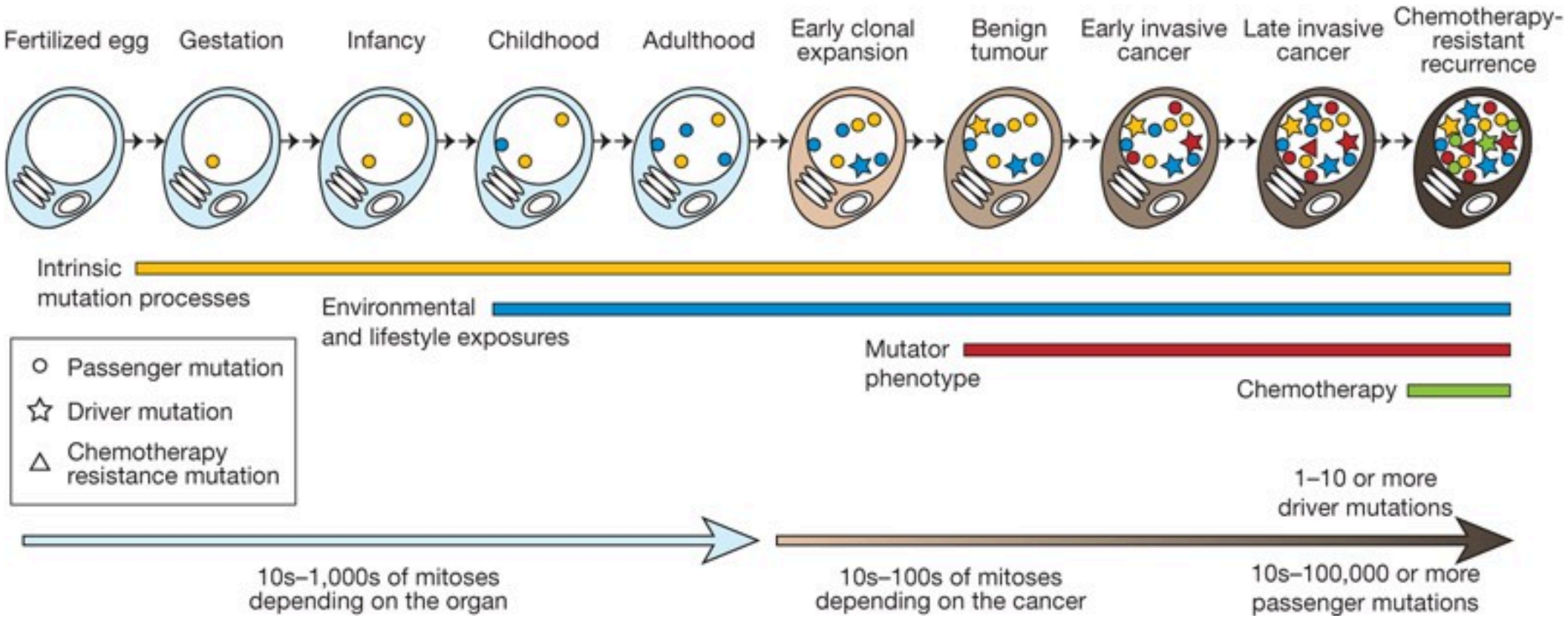


Figure 3: Cancer somatic mutations, The cancer genome, Stratton et al, 2009

Concept of driver mutation

Some mutations are more important than others for tumor progression. Perhaps they are more disruptive and detrimental for the cell to harbor. Those can be observed in multiple cancer types (think of TP53 or BRCA1/2).

> Not all the driver mutations are known and not all the driver mutations are really always drivers.

What Defines a Driver Mutation?

> Not all mutations are driving the tumorigenesis!

- Functional Impact:
  - Driver mutations confer a selective growth advantage to cells, promoting tumor development and progression. These mutations often affect genes regulating cell cycle, apoptosis, DNA repair, and immune evasion.
  - Example pathways: MAPK, PI3K/AKT, and WNT signaling.
- Recurrent Patterns Across Tumors:
  - Frequently observed across different cancer types (e.g., TP53 in over 50% of cancers).
  - Some drivers are specific to tissue types (e.g., EGFR in lung cancer, KIT in gastrointestinal stromal tumors, IDH1 in gliomas).

Types of Driver Mutations

- Oncogenes:
  - Gain-of-function mutations in genes like KRAS, BRAF, and EGFR drive tumor growth by promoting uncontrolled cell division or survival.
- Tumor Suppressors:
  - Loss-of-function mutations in genes like TP53, RB1, and PTEN impair cellular mechanisms that prevent tumor formation.
- Mutator Genes:
  - Mutations in genes like MLH1 or MSH2 lead to genomic instability, enabling the accumulation of additional mutations.

How can we identify what's a mutation's role?

- Can we somehow quantify its importance?

VEP (Variant Effect Predictor)



(a) Ensembl-VEP

What is VEP?

- 2010: The first version of VEP was introduced as part of the Ensembl project.
- 2012-2015 Functional prediction scores were introduced.
- 2017 Other databases are now part of the VEP, such as COSMIC, ClinVar, gnomAD. The tool was made more generalizable and personalizable with the addition of VEP's plugin feature.
- VEP is 100% written in perl VEP Github

Important Annotation Databases

Cancer-Specific Resources

COSMIC (Catalogue of Somatic Mutations in Cancer)

Field	Content
Purpose	Somatic mutations in cancer
Example	`COSV59384583; OCCURENCE=1(skin)`
Interpretation	Mutation seen in skin cancer; frequency indicates driver likelihood

ClinVar (Clinical Variant Database)

Field	Content
Purpose	Clinical significance assessment
Example	`Pathogenic/Likely_pathogenic`
Interpretation	Strong evidence for disease causation

Population & Prediction Resources

gnomAD (Genome Aggregation Database)

Field	Content
Purpose	Population allele frequency
Example	`AF=0.00001`
Interpretation	Very rare (<0.01% frequency)

AlphaMissense

Field	Content
Purpose	AI-predicted functional impact
Example	`likely_pathogenic`
Interpretation	Model predicts damaging effect

Filtering Strategy

Combine databases for variant prioritization:

- **gnomAD AF < 0.01** → Filters common variants
- **COSMIC presence** → Supports somatic driver role
- **ClinVar pathogenic** → Clinical validation
- **AlphaMissense** → Functional prediction for novel variants

What Does VEP Do?

The Challenge: Imagine annotating thousands of variants manually - searching databases, reviewing literature, interpreting functional impact. This would take **weeks** and be highly **error-prone**.

VEP automates annotation to accelerate discovery:

- Annotates genomic variants (somatic and germline)
- Predicts functional consequences
- Integrates multiple databases (COSMIC, ClinVar, gnomAD, etc.)
- Essential for cancer genomics and clinical applications

⚠ **What VEP Does NOT Do**

VEP is powerful, but **it's not magic:**

- **Does NOT call variants** - requires pre-called VCF input
- **Does NOT guarantee accuracy** - annotation quality depends on input quality
- **Does NOT replace human judgment** - false positives/negatives require expert review

Remember: Automation simplifies the process, but critical thinking and validation remain essential!

What is not well understood?

There are no databases investigating those aspects of biology that might play an important role in the cancer development. After all it is estimated that >97% of all the mutations are “passenger events” and they do not have direct impact on the tumor growth.

- The effects of epigenetic changes
- Glycosilations
- Transposomes (like LINE-1)
- SV (inversion for example)
- Alternative splicing impacts
- Non-coding RNAs and their alterations
- Tumor Microenvironment
- Microbiome
- Etc.

In summary: Why do we use VEP?

We can answer questions like:

- How damaging is a certain somatic mutation?
- What is the impact in a particular cancer type?
- Is the mutation known for that cancer?
- Are there therapeutic implications?

Variant Annotation with VEP

VEP: Transcript Selection & Priority

The Challenge: A single variant can have 4-12+ annotations due to multiple transcript isoforms. Which one matters?

1. Handling Transcripts

VEP options to simplify output when a gene has multiple transcripts:

- `--pick`: One consequence per variant (Best for simple filtering)
- `--pick_allele`: One per variant allele
- `--per_gene`: One per gene
- `--flag_pick`: Flags the “chosen” one but keeps others
- `--pick_allele_gene`: Most comprehensive filtering

Selection hierarchy:

1. Canonical transcript (MANE Select for human)
2. Protein coding biotype
3. Most severe consequence
4. Longest transcript

2. Consequence Severity

Top 10 Most Severe:

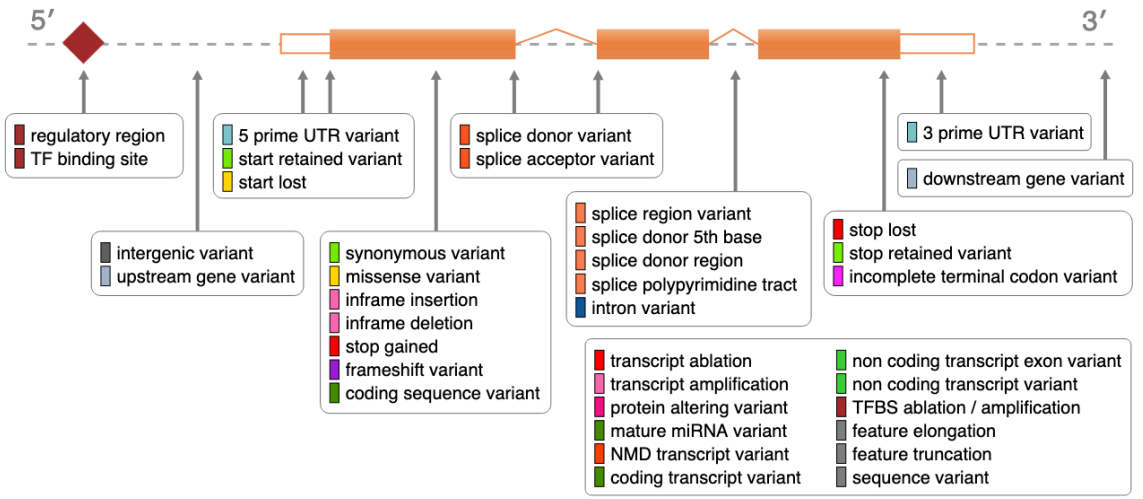
1. **Transcript ablation**
2. **Splice acceptor/donor variant**
3. **Stop gained** (Nonsense)
4. **Frameshift variant**
5. Stop lost
6. Start lost
7. Transcript amplification
8. Inframe insertion/deletion
9. **Missense variant**
10. Protein altering variant

Cancer examples:

- HIGH impact: *TP53* frameshift → TSG loss
- MODERATE impact: *KRAS* p.G12D → oncogene gain

Practical Tip

VEP Impact Categories



VEP's consequences

Impact	Description	Examples
HIGH	**Disruptive.** Likely loss of function.	Frameshift, Stop Gained, Splice acceptor/donor.
MODERATE	**Non-disruptive** change to protein.	Missense, In-frame indel.
LOW	Unlikely to change function.	Synonymous, Splice region (non-canonical).
MODIFIER	Non-coding / Regulatory.	Intronic, UTRs, Intergenic.

⚠ **Important Exceptions**

Not all HIGH impact = pathogenic

Not all MODERATE impact = benign (e.g., *IDH1* p.R132H is LOW but a critical driver mutation)

Always consider biological context!

PolyPhen-2 Score

Purpose: Predicts if a missense mutation is damaging based on protein structure and evolutionary conservation.

Scoring Categories:

1. **Probably Damaging** (> 0.85)
    - *High confidence* functional effect.
2. **Possibly Damaging** (0.15 - 0.85)
    - *Moderate confidence* effect.
3. **Benign** (< 0.15)
    - Likely no phenotypic effect.

> **Usage**
>
> PolyPhen-2 is specific to **human proteins**.
>
> It should be used in conjunction with other tools (like SIFT) to build a consensus on missense variant interpretation.

SIFT Score

Purpose: Predicts if an amino acid substitution is deleterious based on sequence conservation across species.

Scoring Categories:

1. **Deleterious** (≤ 0.05)
    - Substitution at highly conserved position
    - *High confidence* damaging effect
2. **Tolerated** (> 0.05)
    - Substitution at variable position
    - Likely benign or neutral effect

Note: Lower scores = More damaging (opposite to PolyPhen-2!)

Usage

SIFT is available for **10 species** in Ensembl (including human, mouse, zebrafish).

All possible amino acid substitutions are **pre-calculated**, making annotation very fast.

Use in combination with PolyPhen-2 for consensus prediction.

Interpretation Guide

SIFT	PolyPhen-2	Confidence
Deleterious	Probably Damaging	**High** - prioritize
Tolerated	Benign	**High** - likely benign
Deleterious	Benign	Mixed - manual review
Tolerated	Damaging	Mixed - manual review

VEP Plugins

Essential plugins for cancer analysis:

REVEL

```
--plugin REVEL,/path/to/revel/revel_all_chromosomes.tsv.gz
```

> REVEL paper: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants

You can download here the data (~6.5GB): https://sites.google.com/site/revelgenomics/downloads or https://zenodo.org/records/7072866

```
unzip revel-v1.3_all_chromosomes.zip
cat revel_with_transcript_ids | tr "," "\t" > tabbed_revel.tsv
sed '1s/.*/#&/' tabbed_revel.tsv > new_tabbed_revel.tsv
bgzip new_tabbed_revel.tsv
```

Prepare for GRCh38

```
zcat new_tabbed_revel.tsv.gz | head -n1 > h
zgrep -h -v ^#chr new_tabbed_revel.tsv.gz | awk '$3 != "." ' | sort -k1,1 -k3,3n - | cat h - | bgzip -c > new_tabbed_revel_grch38.tsv.gz
tabix -f -s 1 -b 3 -e 3 new_tabbed_revel_grch38.tsv.gz
```

Usage:

```
--plugin REVEL,file=/path/to/revel/data.tsv.gz
```

AlphaMissense

> AlphaMissense's Paper: Accurate proteome-wide missense variant effect prediction with AlphaMissense

Download link

Prepare for GRCh38

```
wget https://storage.googleapis.com/dm_alphamissense/AlphaMissense_hg38.tsv.gz
tabix -s 1 -b 2 -e 2 -f -S 1 AlphaMissense_hg38.tsv.gz
```

Run it with VEP

```
--plugin AlphaMissense,file=/full/path/to/file.tsv.gz
```

dbNSFP

> dbNSFP v4 paper: a comprehensive database of transcript-specific functional predictions and annotations for human nonsynonymous and splice-site SNVs

A VEP plugin that retrieves data for missense variants from a tabix-indexed dbNSFP file.

```
--plugin dbNSFP,/path/to/dbNSFP.gz,SIFT_score,HDIV_score
```

Prepare the data

```
version=4.7c
wget ftp://dbnsfp:dbnsfp@dbnsfp.softgenetics.com/dbNSFP${version}.zip
unzip dbNSFP${version}.zip
zcat dbNSFP${version}_variant.chr1.gz | head -n1 > h
```

Prepare for GRCh38

```
zgrep -h -v ^#chr dbNSFP${version}_variant.chr* | sort -k1,1 -k2,2n - | cat h - | bgzip -c > dbNSFP${version}_grch38.gz
  tabix -s 1 -b 2 -e 2 dbNSFP${version}_grch38.gz
```

Run it with VEP

```
--plugin dbNSFP,/path/to/dbNSFP.gz,LRT_score,GERP++_RS
```

Filter Variants with VEP

You can use the tool that is included in the VEP's suite of tools. This tool generally works very well with data that have been VEP-annotated

Filter SIFT deleterious events

```
filter_vep -i variant_effect_output.txt -filter "SIFT is deleterious" | grep -v "##" | head -n5
```

Can be used with pipes, for example (might save memory)

```
vep -i examples/homo_sapiens_GRCh38.vcf --cache --force_overwrite --sift b --canonical --symbol --tab --fields Uploaded_variation,SYMBOL,CANONICAL,SIFT -o STDOUT
filter_vep --filter "CANONICAL is YES and SIFT is deleterious"
```

Operators:

- `is`: Exact match
- `match`: Regex pattern
- `<`, `>`, `<=`, `>=`: Numeric comparison
- `in`: List membership
- `and`, `or`: Combine conditions

Example 1: High-confidence damaging

```
filter_vep -i vep_output.txt \
  -filter "SIFT is deleterious and \
           PolyPhen is probably_damaging"
```

Notes

- There are many other plugins that can be used depending on the context and the specific biological question at hand. You can have a look here: Plugins for VEP
- You can run multiple plugins at the same time
- The more plugins the more computationally expensive it could become
- There is a nice `help` included in VEP that can be useful for consultation

AI tools to know

- DeepSomatic: deep learning somatic caller, an alternative to Mutect2. It writes a standard VCF, so VEP works on its output too.
- AlphaGenome: predicts the effect of variants outside coding sequence. It is not a caller and not a VEP plugin: you run it from Python.

Best Practices

Quality Control

- Filter low-quality variants before annotation
- Use matched normal samples when available
- Consider sequencing artifacts
- Document filtering criteria
- Use IGV for confirmation

Annotation Strategy

1. Use multiple prediction algorithms
2. Consider tissue-specific expression
3. Include population frequencies
4. Add clinical annotations
5. Follow standardized guidelines

Points to keep in mind

Include MODERATE impact variants

- HIGH impact often affects tumor suppressors (*TP53*, *PTEN*, *RB1*)
- MODERATE impact often affects oncogenes (*KRAS*, *BRAF*, *PIK3CA*)
- Examples: *KRAS* p.G12D, *BRAF* p.V600E are MODERATE but critical drivers

Many drivers remain unknown

- Only ~500-600 cancer genes well-characterized
- Absence from databases ≠ passenger mutation
- Look for recurrent mutations across samples

CNV calling limitations

- Particularly challenging in WES (only 2% of genome covered)
- Requires adequate tumor purity (>40% ideal) and coverage
- VEP is not specialized for CNV annotation (CNVannotator, ClassifyCNV)

> ⚠ **No Tool is Perfect**
>
> Always validate important findings visually (IGV) and consider orthogonal methods for clinical decisions.

Resources

Useful Links

- VEP Documentation
- gnomAD Browser
- COSMIC Database
- ClinVar
- OncoKB

Exercises

- Variant annotation: annotate the Mutect2 calls with VEP
- Visualization: load the VEP output in R and plot it with ggplot2

Questions?

Feel free to drop a line in the chat or to contact us.

Contact Information

- VEP Support: https://www.ensembl.org/Help/Contact
- Documentation: http://www.ensembl.org/info/docs/tools/vep/
- GitHub: https://github.com/Ensembl/ensembl-vep
- Bioconductor: https://bioconductor.org/

Exercises



Giphy